

Supplementary Material: Detailed Appendices

Accompanying the doctoral thesis

Multiscale Modelling of Weakly Porous Grid Flows

ANR MultiMatchGrid Project

Hossameldin ABDELAZIZ

Supervised by Associate Professor Francesco Romanò

Laboratoire de Mécanique des Fluides de Lille (LMFL)

Arts et Métiers Institute of Technology, Lille, France

2026

List of Figures

B.1	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 0.05$	5
B.2	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 0.05$	6
B.3	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 0.1$	7
B.4	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 0.1$	8
B.5	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 0.5$	9
B.6	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 0.5$	10
B.7	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 1$	11
B.8	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 1$	12
B.9	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 5$	13
B.10	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 5$	14
B.11	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 10$	15
B.12	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 10$	16
B.13	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 20$	17
B.14	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 20$	18
B.15	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 33.33$	19
B.16	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 33.33$	20
B.17	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 50$	21
B.18	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 50$	22
B.19	Streamwise evolution of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 100$	23
B.20	Streamwise evolution of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 100$	24
B.21	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 0.05$	26
B.22	Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 0.05$	27
B.23	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 0.1$	28
B.24	Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 0.1$	29
B.25	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 0.5$	30
B.26	Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 0.5$	31
B.27	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 1$	32
B.28	Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 1$	33
B.29	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 5$	34
B.30	Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 5$	35
B.31	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 10$	36
B.32	Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 10$	37
B.33	Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 20$	38

B.34 Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 20$	39
B.35 Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 33.33$	40
B.36 Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 33.33$	41
B.37 Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 50$	42
B.38 Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 50$	43
B.39 Streamwise evolution of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 100$	44
B.40 Streamwise evolution of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 100$	45
B.41 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 0.05$	47
B.42 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 0.05$	48
B.43 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 0.1$	49
B.44 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 0.1$	50
B.45 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 0.5$	51
B.46 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 0.5$	52
B.47 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 1$	53
B.48 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 1$	54
B.49 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 5$	55
B.50 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 5$	56
B.51 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 10$	57
B.52 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 10$	58
B.53 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 20$	59
B.54 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 20$	60
B.55 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 33.33$	61
B.56 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 33.33$	62
B.57 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 50$	63
B.58 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 50$	64
B.59 Streamwise evolution of $\Re(\hat{U}_{m,n})$ at $\text{Re}_H = 100$	65
B.60 Streamwise evolution of $\Im(\hat{U}_{m,n})$ at $\text{Re}_H = 100$	66
B.61 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 0.05$.	68
B.62 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 0.1$.	69
B.63 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 0.5$.	70
B.64 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 1$. .	71
B.65 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 5$. .	72
B.66 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 10$. .	73
B.67 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 20$. .	74
B.68 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 33.33$	75
B.69 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 50$. .	76
B.70 Streamwise evolution of normalized SRG modal energy at $\text{Re}_H = 100$.	77
B.71 TIDFT reconstruction at $\phi = 0.1$, $\text{Re}_H = 0.05$ and 0.1	79
B.72 TIDFT reconstruction at $\phi = 0.1$, $\text{Re}_H = 0.5$ and 1	80
B.73 TIDFT reconstruction at $\phi = 0.1$, $\text{Re}_H = 5$ and 10	81
B.74 TIDFT reconstruction at $\phi = 0.1$, $\text{Re}_H = 20$ and 33.33	82
B.75 TIDFT reconstruction at $\phi = 0.1$, $\text{Re}_H = 50$ and 100	83
B.76 TIDFT reconstruction at $\phi = 0.15$, $\text{Re}_H = 0.05$ and 0.1	84
B.77 TIDFT reconstruction at $\phi = 0.15$, $\text{Re}_H = 0.5$ and 1	85
B.78 TIDFT reconstruction at $\phi = 0.15$, $\text{Re}_H = 5$ and 10	86

B.79	TIDFT reconstruction at $\phi = 0.15$, $\text{Re}_H = 20$ and 33.33	87
B.80	TIDFT reconstruction at $\phi = 0.15$, $\text{Re}_H = 50$ and 100	88
B.81	TIDFT reconstruction at $\phi = 0.2$, $\text{Re}_H = 0.05$ and 0.1	89
B.82	TIDFT reconstruction at $\phi = 0.2$, $\text{Re}_H = 0.5$ and 1	90
B.83	TIDFT reconstruction at $\phi = 0.2$, $\text{Re}_H = 5$ and 10	91
B.84	TIDFT reconstruction at $\phi = 0.2$, $\text{Re}_H = 20$ and 33.33	92
B.85	TIDFT reconstruction at $\phi = 0.2$, $\text{Re}_H = 50$ and 100	93
B.86	TIDFT reconstruction at $\phi = 0.25$, $\text{Re}_H = 0.05$ and 0.1	94
B.87	TIDFT reconstruction at $\phi = 0.25$, $\text{Re}_H = 0.5$ and 1	95
B.88	TIDFT reconstruction at $\phi = 0.25$, $\text{Re}_H = 5$ and 10	96
B.89	TIDFT reconstruction at $\phi = 0.25$, $\text{Re}_H = 20$ and 33.33	97
B.90	TIDFT reconstruction at $\phi = 0.25$, $\text{Re}_H = 50$ and 100	98
B.91	Relative TIDFT reconstruction error for streamwise velocity at $\phi = 0.1$	100
B.92	Relative TIDFT reconstruction error for streamwise velocity at $\phi = 0.15$	101
B.93	Relative TIDFT reconstruction error for streamwise velocity at $\phi = 0.2$	102
B.94	Relative TIDFT reconstruction error for streamwise velocity at $\phi = 0.25$	103
B.95	Relative TIDFT reconstruction error for transverse velocity at $\phi = 0.1$	104
B.96	Relative TIDFT reconstruction error for transverse velocity at $\phi = 0.15$	105
B.97	Relative TIDFT reconstruction error for transverse velocity at $\phi = 0.2$	106
B.98	Relative TIDFT reconstruction error for transverse velocity at $\phi = 0.25$	107
B.99	Relative TIDFT reconstruction error for spanwise velocity at $\phi = 0.1$	108
B.100	Relative TIDFT reconstruction error for spanwise velocity at $\phi = 0.15$	109
B.101	Relative TIDFT reconstruction error for spanwise velocity at $\phi = 0.2$	110
B.102	Relative TIDFT reconstruction error for spanwise velocity at $\phi = 0.25$	111
B.103	Relative TIDFT reconstruction error for streamwise force at $\phi = 0.1$	112
B.104	Relative TIDFT reconstruction error for streamwise force at $\phi = 0.15$	113
B.105	Relative TIDFT reconstruction error for streamwise force at $\phi = 0.2$	114
B.106	Relative TIDFT reconstruction error for streamwise force at $\phi = 0.25$	115
C.1	Mode-wise neural-network regressor for predicting Fourier coefficients	117
C.2	Reference DFT and ML prediction of $\Re(\hat{W}_{0,1})$	118
C.3	Reference DFT and ML prediction of $\Im(\hat{W}_{0,1})$	119
C.4	Reference DFT and ML prediction of $\Re(\hat{W}_{0,2})$	120
C.5	Reference DFT and ML prediction of $\Im(\hat{W}_{0,2})$	121
C.6	Reference DFT and ML prediction of $\Re(\hat{W}_{0,3})$	122
C.7	Reference DFT and ML prediction of $\Im(\hat{W}_{0,3})$	123
C.8	Reference DFT and ML prediction of $\Re(\hat{W}_{1,1})$	124
C.9	Reference DFT and ML prediction of $\Im(\hat{W}_{1,1})$	125
C.10	Reference DFT and ML prediction of $\Re(\hat{W}_{1,2})$	126
C.11	Reference DFT and ML prediction of $\Im(\hat{W}_{1,2})$	127
C.12	Reference DFT and ML prediction of $\Re(\hat{W}_{1,3})$	128
C.13	Reference DFT and ML prediction of $\Im(\hat{W}_{1,3})$	129
C.14	Reference DFT and ML prediction of $\Re(\hat{W}_{2,2})$	130
C.15	Reference DFT and ML prediction of $\Im(\hat{W}_{2,2})$	131
C.16	Reference DFT and ML prediction of $\Re(\hat{W}_{2,3})$	132
C.17	Reference DFT and ML prediction of $\Im(\hat{W}_{2,3})$	133

C.18 Reference DFT and ML prediction of $\Re(\hat{W}_{3,3})$	134
C.19 Reference DFT and ML prediction of $\Im(\hat{W}_{3,3})$	135
C.20 Reference DFT and ML prediction of $\Re(\hat{V}_{0,1})$	136
C.21 Reference DFT and ML prediction of $\Im(\hat{V}_{0,1})$	137
C.22 Reference DFT and ML prediction of $\Re(\hat{V}_{0,2})$	138
C.23 Reference DFT and ML prediction of $\Im(\hat{V}_{0,2})$	139
C.24 Reference DFT and ML prediction of $\Re(\hat{V}_{0,3})$	140
C.25 Reference DFT and ML prediction of $\Im(\hat{V}_{0,3})$	141
C.26 Reference DFT and ML prediction of $\Re(\hat{V}_{1,1})$	142
C.27 Reference DFT and ML prediction of $\Im(\hat{V}_{1,1})$	143
C.28 Reference DFT and ML prediction of $\Re(\hat{V}_{1,2})$	144
C.29 Reference DFT and ML prediction of $\Im(\hat{V}_{1,2})$	145
C.30 Reference DFT and ML prediction of $\Re(\hat{V}_{1,3})$	146
C.31 Reference DFT and ML prediction of $\Im(\hat{V}_{1,3})$	147
C.32 Reference DFT and ML prediction of $\Re(\hat{V}_{2,1})$	148
C.33 Reference DFT and ML prediction of $\Im(\hat{V}_{2,1})$	149
C.34 Reference DFT and ML prediction of $\Re(\hat{V}_{2,2})$	150
C.35 Reference DFT and ML prediction of $\Im(\hat{V}_{2,2})$	151
C.36 Reference DFT and ML prediction of $\Re(\hat{V}_{2,3})$	152
C.37 Reference DFT and ML prediction of $\Im(\hat{V}_{2,3})$	153
C.38 Reference DFT and ML prediction of $\Re(\hat{V}_{3,1})$	154
C.39 Reference DFT and ML prediction of $\Im(\hat{V}_{3,1})$	155
C.40 Reference DFT and ML prediction of $\Re(\hat{V}_{3,2})$	156
C.41 Reference DFT and ML prediction of $\Im(\hat{V}_{3,2})$	157
C.42 Reference DFT and ML prediction of $\Re(\hat{V}_{3,3})$	158
C.43 Reference DFT and ML prediction of $\Im(\hat{V}_{3,3})$	159
C.44 DFT and ML comparison of $\Re(\hat{W}_{m,n})$ at $\text{Re}_H = 100$	160
C.45 DFT and ML comparison of $\Im(\hat{W}_{m,n})$ at $\text{Re}_H = 100$	161
C.46 DFT and ML comparison of $\Re(\hat{V}_{m,n})$ at $\text{Re}_H = 100$	162
C.47 DFT and ML comparison of $\Im(\hat{V}_{m,n})$ at $\text{Re}_H = 100$	163
C.48 Training and validation MSE for $\hat{W}_{m,n}$	164
C.49 Training and validation MSE for $\hat{V}_{m,n}$	165
D.1 Geometry-resolved SSF and VCDFM configurations	166
D.2 Validity map of the VCDFM for the 1×1 SRG	168
D.3 Transverse velocity: SSF versus SSF-VCDFM at $\phi = 0.10$	169
D.4 Transverse velocity: SSF versus TIDFT-VCDFM at $\phi = 0.10$	170
D.5 Transverse velocity: SSF versus ML-VCDFM at $\phi = 0.10$	171
D.6 Transverse velocity: SSF versus SSF-VCDFM at $\phi = 0.15$	172
D.7 Transverse velocity: SSF versus TIDFT-VCDFM at $\phi = 0.15$	173
D.8 Transverse velocity: SSF versus ML-VCDFM at $\phi = 0.15$	174
D.9 Transverse velocity: SSF versus SSF-VCDFM at $\phi = 0.20$	175
D.10 Transverse velocity: SSF versus TIDFT-VCDFM at $\phi = 0.20$	176
D.11 Transverse velocity: SSF versus ML-VCDFM at $\phi = 0.20$	177
D.12 Transverse velocity: SSF versus SSF-VCDFM at $\phi = 0.25$	178

D.13 Transverse velocity: SSF versus TIDFT-VCDFM at $\phi = 0.25$	179
D.14 Transverse velocity: SSF versus ML-VCDFM at $\phi = 0.25$	180
D.15 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 0.05$	182
D.16 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML- VCDFM at $\phi = 0.1$, $Re_H = 0.05$	183
D.17 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 0.05$	184
D.18 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.05$	185
D.19 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.05$	186
D.20 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.05$	187
D.21 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 0.1$	188
D.22 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML- VCDFM at $\phi = 0.1$, $Re_H = 0.1$	189
D.23 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 0.1$	190
D.24 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.1$	191
D.25 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.1$	192
D.26 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.1$	193
D.27 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 0.5$	194
D.28 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML- VCDFM at $\phi = 0.1$, $Re_H = 0.5$	195
D.29 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 0.5$	196
D.30 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.5$	197
D.31 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.5$	198
D.32 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 0.5$	199
D.33 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 1$	200
D.34 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML- VCDFM at $\phi = 0.1$, $Re_H = 1$	201
D.35 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 1$	202
D.36 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 1$	203
D.37 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 1$	204
D.38 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 1$	205
D.39 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 5$	206
D.40 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML- VCDFM at $\phi = 0.1$, $Re_H = 5$	207
D.41 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 5$	208
D.42 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 5$	209
D.43 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 5$	210
D.44 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 5$	211

D.45 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 10$	212
D.46 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 10$	213
D.47 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 10$	214
D.48 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 10$.	215
D.49 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 10$	216
D.50 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 10$.	217
D.51 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 20$	218
D.52 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 20$	219
D.53 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 20$	220
D.54 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 20$.	221
D.55 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 20$	222
D.56 Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 20$.	223
D.57 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.05$	224
D.58 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.05$	225
D.59 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.05$	226
D.60 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.05$	227
D.61 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.05$	228
D.62 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.05$	229
D.63 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.1$	230
D.64 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.1$	231
D.65 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.1$	232
D.66 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.1$	233
D.67 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.1$	234
D.68 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.1$	235
D.69 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.5$	236
D.70 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.5$	237
D.71 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 0.5$	238
D.72 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.5$	239
D.73 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.5$	240
D.74 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 0.5$	241

D.75 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 1$	242
D.76 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 1$	243
D.77 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 1$	244
D.78 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 1$.	245
D.79 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 1$	246
D.80 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 1$.	247
D.81 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 5$	248
D.82 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 5$	249
D.83 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 5$	250
D.84 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 5$.	251
D.85 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 5$	252
D.86 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 5$.	253
D.87 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 10$	254
D.88 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 10$	255
D.89 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 10$	256
D.90 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 10$	257
D.91 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 10$	258
D.92 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 10$.	259
D.93 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 20$	260
D.94 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 20$	261
D.95 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 20$	262
D.96 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 20$	263
D.97 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 20$	264
D.98 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 20$.	265
D.99 Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 33.33$	266
D.100 Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 33.33$	267
D.101 Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 33.33$	268
D.102 Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 33.33$	269
D.103 Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 33.33$	270
D.104 Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 33.33$	271

D.105	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.05$	272
D.106	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.05$	273
D.107	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.05$	274
D.108	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.05$	275
D.109	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.05$	276
D.110	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.05$	277
D.111	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.1$	278
D.112	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.1$	279
D.113	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.1$	280
D.114	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.1$	281
D.115	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.1$	282
D.116	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.1$	283
D.117	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.5$	284
D.118	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.5$	285
D.119	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 0.5$	286
D.120	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.5$	287
D.121	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.5$	288
D.122	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $Re_H = 0.5$	289
D.123	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 1$	290
D.124	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 1$	291
D.125	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 1$	292
D.126	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $Re_H = 1$	293
D.127	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $Re_H = 1$	294
D.128	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $Re_H = 1$	295
D.129	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 5$	296
D.130	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 5$	297
D.131	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 5$	298
D.132	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $Re_H = 5$	299
D.133	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $Re_H = 5$	300
D.134	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $Re_H = 5$	301
D.135	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $Re_H = 10$	302

D.136	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 10$	303
D.137	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 10$	304
D.138	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 10$	305
D.139	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 10$	306
D.140	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 10$	307
D.141	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 20$	308
D.142	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 20$	309
D.143	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 20$	310
D.144	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 20$	311
D.145	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 20$	312
D.146	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 20$	313
D.147	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 33.33$	314
D.148	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 33.33$	315
D.149	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 33.33$	316
D.150	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 33.33$	317
D.151	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 33.33$	318
D.152	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 33.33$	319
D.153	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 50$	320
D.154	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 50$	321
D.155	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 50$	322
D.156	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 50$	323
D.157	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 50$	324
D.158	Detailed SSF and ML-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 50$	325
D.159	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 0.05$	326
D.160	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 0.05$	327
D.161	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 0.05$	328
D.162	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $\text{Re}_H = 0.05$	329
D.163	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $\text{Re}_H = 0.05$	330
D.164	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $\text{Re}_H = 0.05$	331
D.165	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 0.1$	332

D.16	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 0.1$	333
D.167	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 0.1$	334
D.168	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 0.1$	335
D.169	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 0.1$	336
D.170	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 0.1$	337
D.171	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 0.5$	338
D.172	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 0.5$	339
D.173	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 0.5$	340
D.174	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 0.5$	341
D.175	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 0.5$	342
D.176	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 0.5$	343
D.177	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 1$	344
D.178	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 1$	345
D.179	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 1$	346
D.180	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 1$	347
D.181	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 1$	348
D.182	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 1$	349
D.183	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 5$	350
D.184	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 5$	351
D.185	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 5$	352
D.186	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 5$	353
D.187	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 5$	354
D.188	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 5$	355
D.189	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 10$	356
D.190	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 10$	357
D.191	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 10$	358
D.192	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 10$	359
D.193	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 10$	360
D.194	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 10$	361
D.195	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 20$	362
D.196	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 20$	363

D.197	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 20$	364
D.198	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 20$	365
D.199	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 20$	366
D.200	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 20$	367
D.201	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 33.33$	368
D.202	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 33.33$	369
D.203	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 33.33$	370
D.204	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 33.33$	371
D.205	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 33.33$	372
D.206	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 33.33$	373
D.207	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 50$	374
D.208	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 50$	375
D.209	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $Re_H = 50$	376
D.210	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 50$	377
D.211	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 50$	378
D.212	Detailed SSF and ML-VCDFM comparison at $\phi = 0.25$ and $Re_H = 50$	379
D.213	Streamwise comparison of SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.25$, $Re_H = 100$	380
D.214	Flow fields from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.25$, $Re_H = 100$	381
D.215	Velocity profiles from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.25$, $Re_H = 100$	382
D.216	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $Re_H = 100$	383
D.217	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $Re_H = 100$	384
D.218	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 33.33$	385
D.219	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 33.33$	386
D.220	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 33.33$	387
D.221	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 33.33$	388
D.222	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 33.33$	389
D.223	Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 33.33$	390
D.224	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 50$	391
D.225	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 50$	392

D.226	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 50$	393
D.227	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 50$.	394
D.228	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 50$	395
D.229	Detailed SSF and ML-VCDFM comparison at $\phi = 0.1$ and $Re_H = 50$.	396
D.230	Streamwise comparison of SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.1$, $Re_H = 100$	397
D.231	Flow fields from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.1$, $Re_H = 100$	398
D.232	Velocity profiles from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.1$, $Re_H = 100$	399
D.233	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 100$	400
D.234	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 100$	401
D.235	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 50$	402
D.236	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 50$	403
D.237	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $Re_H = 50$	404
D.238	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 50$	405
D.239	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 50$	406
D.240	Detailed SSF and ML-VCDFM comparison at $\phi = 0.15$ and $Re_H = 50$.	407
D.241	Streamwise comparison of SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.15$, $Re_H = 100$	408
D.242	Flow fields from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.15$, $Re_H = 100$	409
D.243	Velocity profiles from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.15$, $Re_H = 100$	410
D.244	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $Re_H = 100$	411
D.245	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $Re_H = 100$	412
D.246	Streamwise comparison of SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.2$, $Re_H = 100$	413
D.247	Flow fields from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.2$, $Re_H = 100$	414
D.248	Velocity profiles from SSF, DFM, SSF-VCDFM, and TIDFT-VCDFM at $\phi = 0.2$, $Re_H = 100$	415
D.249	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $Re_H = 100$	416
D.250	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $Re_H = 100$	417
D.251	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 100$	419
D.252	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 100$	420
D.253	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.1$, $Re_H = 100$	421
D.254	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.1$ and $Re_H = 100$	422
D.255	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.1$ and $Re_H = 100$	423

D.256	Detailed SSF and ML-VCDFM extrapolation comparison at $\phi = 0.1$ and $\text{Re}_H = 100$	424
D.257	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $\text{Re}_H = 100$	425
D.258	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $\text{Re}_H = 100$	426
D.259	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.15$, $\text{Re}_H = 100$	427
D.260	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.15$ and $\text{Re}_H = 100$	428
D.261	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.15$ and $\text{Re}_H = 100$	429
D.262	Detailed SSF and ML-VCDFM extrapolation comparison at $\phi = 0.15$ and $\text{Re}_H = 100$	430
D.263	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 100$	431
D.264	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 100$	432
D.265	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.2$, $\text{Re}_H = 100$	433
D.266	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 100$	434
D.267	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.2$ and $\text{Re}_H = 100$	435
D.268	Detailed SSF and ML-VCDFM extrapolation comparison at $\phi = 0.2$ and $\text{Re}_H = 100$	436
D.269	Streamwise comparison of SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 100$	437
D.270	Flow fields from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 100$	438
D.271	Velocity profiles from SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM at $\phi = 0.25$, $\text{Re}_H = 100$	439
D.272	Detailed SSF and SSF-VCDFM comparison at $\phi = 0.25$ and $\text{Re}_H = 100$	440
D.273	Detailed SSF and TIDFT-VCDFM comparison at $\phi = 0.25$ and $\text{Re}_H = 100$	441
D.274	Detailed SSF and ML-VCDFM extrapolation comparison at $\phi = 0.25$ and $\text{Re}_H = 100$	442

Contents

List of Figures	i
Contents	ii
A Numerical Implementation of the Uni-Directional Model	1
B Detailed Spectral Analysis and Reconstruction Errors	2
C Detailed Machine-Learning Validation of Fourier Coefficients	116
C.1 Prediction within the training parameter range	118
C.2 Extrapolation at $\text{Re}_H = 100$	160
C.3 Training Histories	164
D Detailed Validation of the VCDFM Against the SSF	166
D.1 Effective Validity Range of the VCDFM	167
D.2 Detailed Validation Cases Within the Effective Regime	181
D.3 Detailed Validation Cases in the Breakdown Regime	385
D.4 ML-VCDFM Validation under Reynolds-Number Extrapolation	418

Appendix A

Numerical Implementation of the Uni-Directional Model

To obtain a reduced-order description of the near-grid flow, we solve a two-dimensional Poisson problem for the axial velocity $w_0(x, y)$ over the aperture cross-section Ω :

$$\nabla_{xy}^2 w_0 = C, \quad w_0|_{\partial\Omega} = 0, \quad \int_{\Omega} w_0 dx dy = \cos \alpha, \quad (\text{A.1})$$

where $\partial\Omega$ is the solid boundary and C is a geometry/porosity-dependent constant chosen such that the total flux matches the imposed inflow. This defines our Uni-Directional Model (UDM): it yields the leading-order axial field $w_0(x, y)$ in the Inner region and the associated streamwise pressure gradient $\partial_z p_0 = C(\phi, \text{geometry})$.

Numerically, we implemented a custom OpenFOAM scalar solver (UDM) that advances

$$\partial_t W - \nabla_{xy}^2 W = 1$$

in pseudo-time until steady state, where $W \equiv \hat{w}$ is the unit-forcing solution of $\nabla_{xy}^2 \hat{w} = 1$ with $\hat{w}|_{\partial\Omega} = 0$. No integral flux constraint is imposed during the solve. After convergence, we compute the normalized flux

$$\hat{Q} = \int_{\Omega} \hat{w} dx dy,$$

and rescale to the $Q = 1$ convention used in the Stokes simulations:

$$w_0 = \frac{1}{\hat{Q}} \hat{w}, \quad (\partial_z p_0)_{UDM} = \frac{1}{\hat{Q}}.$$

The computation uses the same finite-volume mesh family as the Stokes runs, restricted to a single grid plane. Spatial discretization is second order (Gauss linear Laplacian). The linear systems are solved with a preconditioned conjugate-gradient (PCG) method and a diagonal incomplete Cholesky (DIC) preconditioner; non-orthogonality corrections are enabled when required by geometry. Convergence is declared when residuals drop below 10^{-9} .

Appendix B

Detailed Spectral Analysis and Reconstruction Errors

This appendix presents a case-by-case assessment of the spectral structure and reconstruction accuracy of the geometry-resolved small-scale flow (SSF). The results concern the square regular grid (SRG) over the porosity and Reynolds-number ranges indicated in the figures. The velocity field is denoted by $\mathbf{u} = (u, v, w)$, where w is the streamwise component and u, v are the transverse components. Lengths are normalized by the grid spacing H , and $\Gamma = t/H$ is the dimensionless grid thickness.

The downstream flow generated by a weakly porous grid depends on the grid geometry, porosity ϕ , and Reynolds number Re_H . To examine its transverse spatial scales, the SSF velocity field is decomposed in the periodic x - and y -directions at successive streamwise locations.

At $z = z_k$, the velocity field sampled on a transverse plane is written as

$$\mathbf{u}_{\text{SSF}}(x_p, y_q; z_k) = (u_{\text{SSF}}, v_{\text{SSF}}, w_{\text{SSF}})(x_p, y_q; z_k), \quad (\text{B.1})$$

where

$$p = 0, \dots, N_x - 1, \quad q = 0, \dots, N_y - 1, \quad k = 0, \dots, N_z - 1.$$

The two-dimensional discrete Fourier coefficients on each plane are

$$\hat{\mathbf{U}}_{m,n}(z_k) = \frac{1}{N_x N_y} \sum_{p=0}^{N_x-1} \sum_{q=0}^{N_y-1} \mathbf{u}_{\text{SSF}}(x_p, y_q; z_k) \exp \left[-2\pi i \left(\frac{mp}{N_x} + \frac{nq}{N_y} \right) \right], \quad (\text{B.2})$$

with $m = 0, \dots, N_x - 1$ and $n = 0, \dots, N_y - 1$. The coefficient vector is

$$\hat{\mathbf{U}}_{m,n} = (\hat{U}_{m,n}, \hat{V}_{m,n}, \hat{W}_{m,n}),$$

corresponding to the velocity components u , v , and w .

The associated discrete transverse wavenumbers are

$$\kappa_m = \begin{cases} \frac{2\pi m}{L_x}, & 0 \leq m < \frac{N_x}{2}, \\ \frac{2\pi(m - N_x)}{L_x}, & \frac{N_x}{2} \leq m < N_x, \end{cases} \quad (\text{B.3a})$$

$$\kappa_n = \begin{cases} \frac{2\pi n}{L_y}, & 0 \leq n < \frac{N_y}{2}, \\ \frac{2\pi(n - N_y)}{L_y}, & \frac{N_y}{2} \leq n < N_y. \end{cases} \quad (\text{B.3b})$$

Here, L_x and L_y are the dimensionless transverse domain lengths. For the periodic unit cell considered here, $L_x = L_y = 1$.

The energy associated with a transverse Fourier mode is defined as

$$E_{m,n}(z_k) = |\hat{U}_{m,n}(z_k)|^2 + |\hat{V}_{m,n}(z_k)|^2 + |\hat{W}_{m,n}(z_k)|^2, \quad (\text{B.4})$$

and the total spectral energy on that plane is

$$E_{\text{tot}}(z_k) = \sum_{m=0}^{N_x-1} \sum_{n=0}^{N_y-1} E_{m,n}(z_k). \quad (\text{B.5})$$

The ratio $E_{m,n}/E_{\text{tot}}$ gives the contribution of each mode to the energy of the full transverse velocity field.

For reconstruction, the full inverse discrete Fourier transform uses all modes:

$$\mathbf{u}_{\text{FIDFT}}(x_p, y_q; z_k) = \Re \left[\sum_{m=0}^{N_x-1} \sum_{n=0}^{N_y-1} \hat{\mathbf{U}}_{m,n}(z_k) \exp \left[2\pi i \left(\frac{mp}{N_x} + \frac{nq}{N_y} \right) \right] \right]. \quad (\text{B.6})$$

The truncated inverse discrete Fourier transform (TIDFT) instead retains the modes $0 \leq m \leq m_T$ and $0 \leq n \leq n_T$ in the non-negative spectral quadrant. For symmetry-preserving flows, their corresponding modes in the remaining quadrants are completed using the Fourier symmetry relations before the inverse transform. The resulting velocity field is denoted by $\mathbf{u}_{\text{TIDFT}}$.

Spatial derivatives needed to reconstruct the streamwise forcing can be evaluated from the Fourier representation. In the periodic directions,

$$\mathcal{F} \left(\frac{\partial \mathbf{u}}{\partial x} \right)_{m,n} = i\kappa_m \hat{\mathbf{U}}_{m,n}, \quad (\text{B.7a})$$

$$\mathcal{F} \left(\frac{\partial \mathbf{u}}{\partial y} \right)_{m,n} = i\kappa_n \hat{\mathbf{U}}_{m,n}, \quad (\text{B.7b})$$

$$\mathcal{F} \left(\frac{\partial^2 \mathbf{u}}{\partial x^2} \right)_{m,n} = -\kappa_m^2 \hat{\mathbf{U}}_{m,n}, \quad (\text{B.7c})$$

$$\mathcal{F} \left(\frac{\partial^2 \mathbf{u}}{\partial y^2} \right)_{m,n} = -\kappa_n^2 \hat{\mathbf{U}}_{m,n}. \quad (\text{B.7d})$$

For equally spaced interior planes, the corresponding streamwise derivatives of a modal coefficient are approximated by

$$\frac{\partial \hat{U}_{m,n}}{\partial z}(z_k) \approx \frac{\hat{U}_{m,n}(z_{k+1}) - \hat{U}_{m,n}(z_{k-1})}{2\Delta z}, \quad (\text{B.8a})$$

$$\frac{\partial^2 \hat{U}_{m,n}}{\partial z^2}(z_k) \approx \frac{\hat{U}_{m,n}(z_{k+1}) - 2\hat{U}_{m,n}(z_k) + \hat{U}_{m,n}(z_{k-1}))}{(\Delta z)^2}. \quad (\text{B.8b})$$

The streamwise forcing reconstructed from a velocity field is

$$F_z = -(\mathbf{u} \cdot \nabla)w - \frac{1}{\text{Re}_H} \nabla^2 w. \quad (\text{B.9})$$

Applying this definition to the geometry-resolved or TIDFT velocity field gives $F_{z,\text{SSF}}$ or $F_{z,\text{TIDFT}}$, respectively.

The first series of figures shows the streamwise evolution of the Fourier coefficients $\hat{W}_{m,n}$ of the streamwise velocity. These plots show how individual modes vary with porosity, Reynolds number, and distance downstream of the grid.

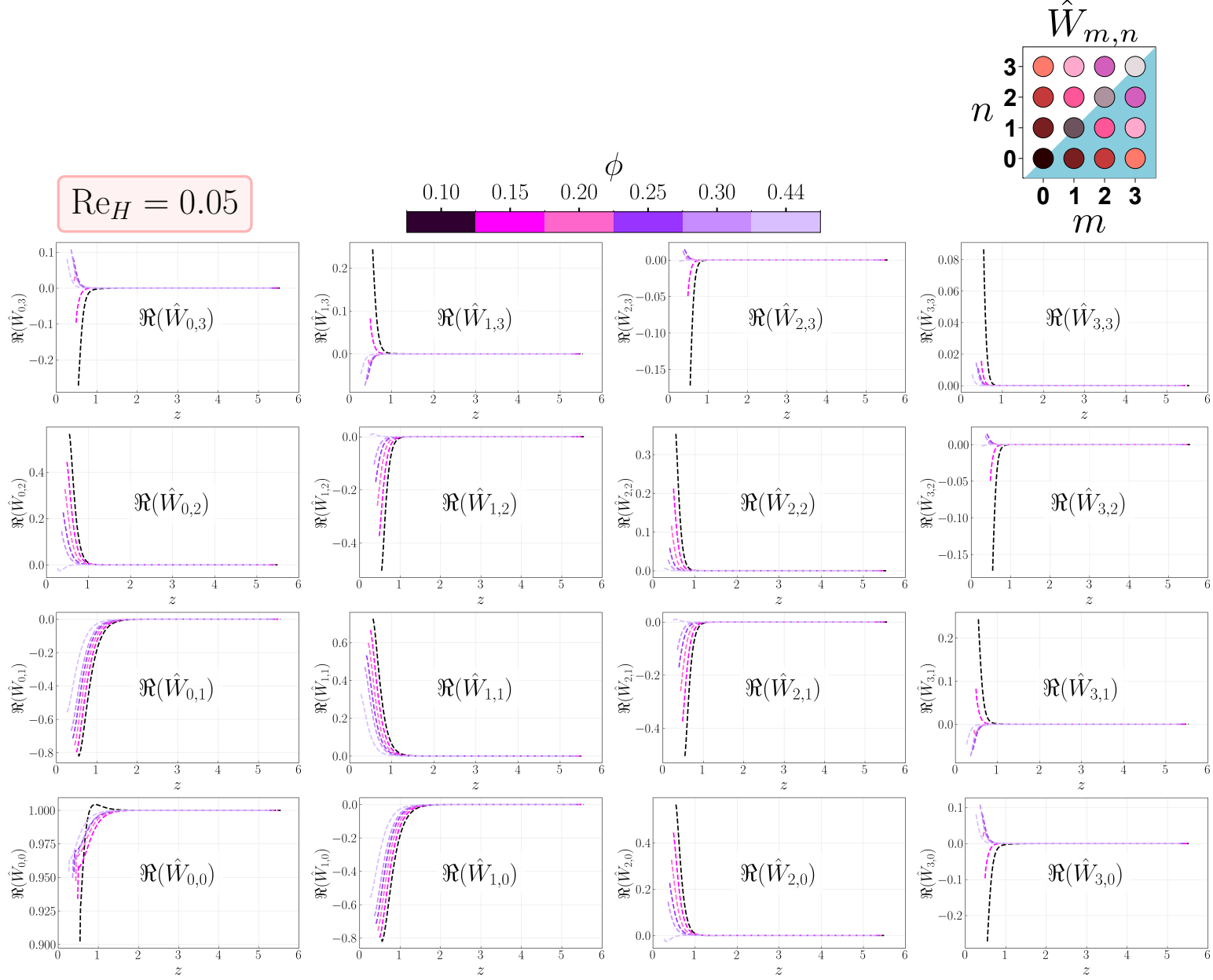


Figure B.1: Streamwise evolution of the real part $\mathcal{R}(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.05$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

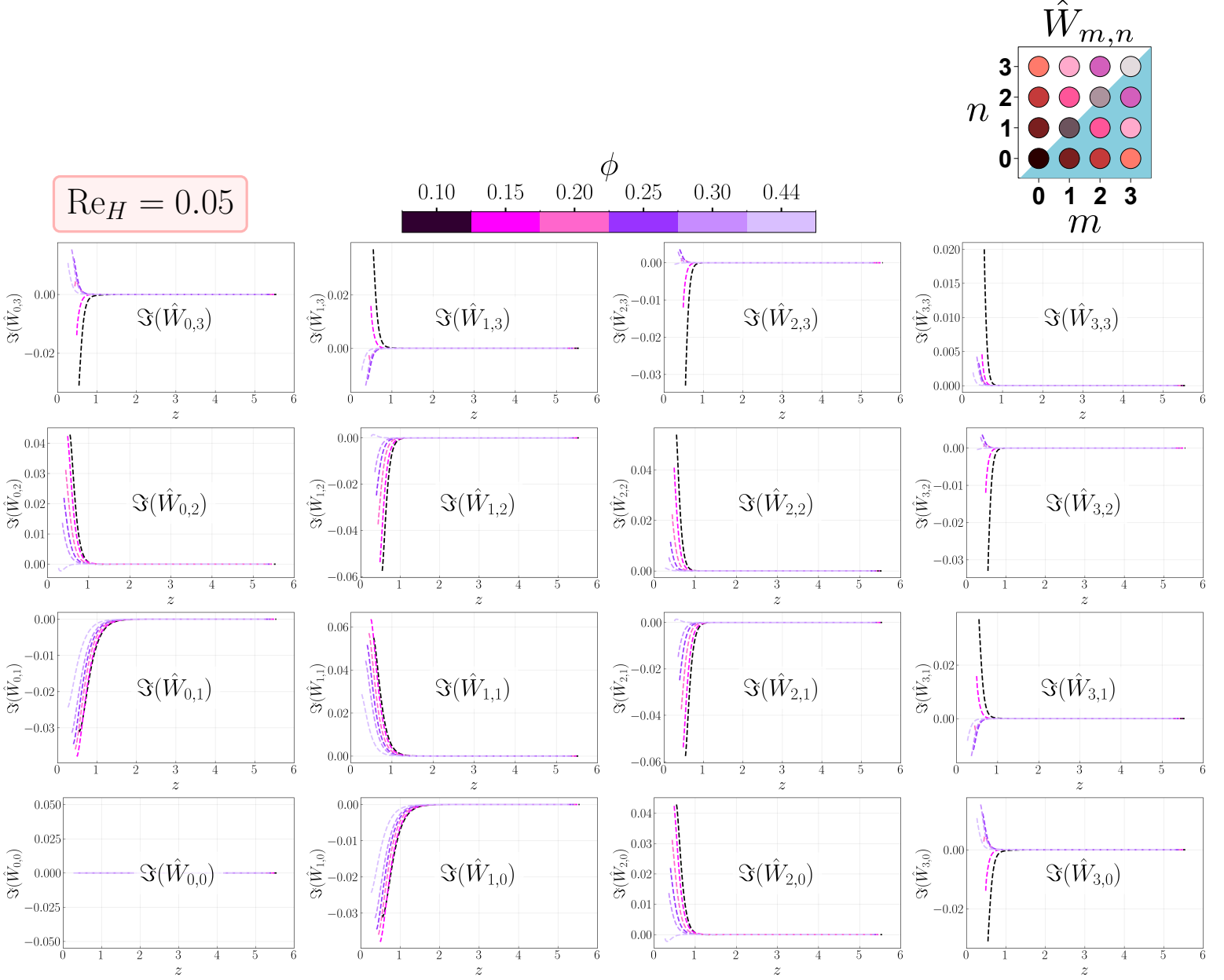


Figure B.2: Streamwise evolution of the imaginary part $\mathfrak{I}(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.05$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

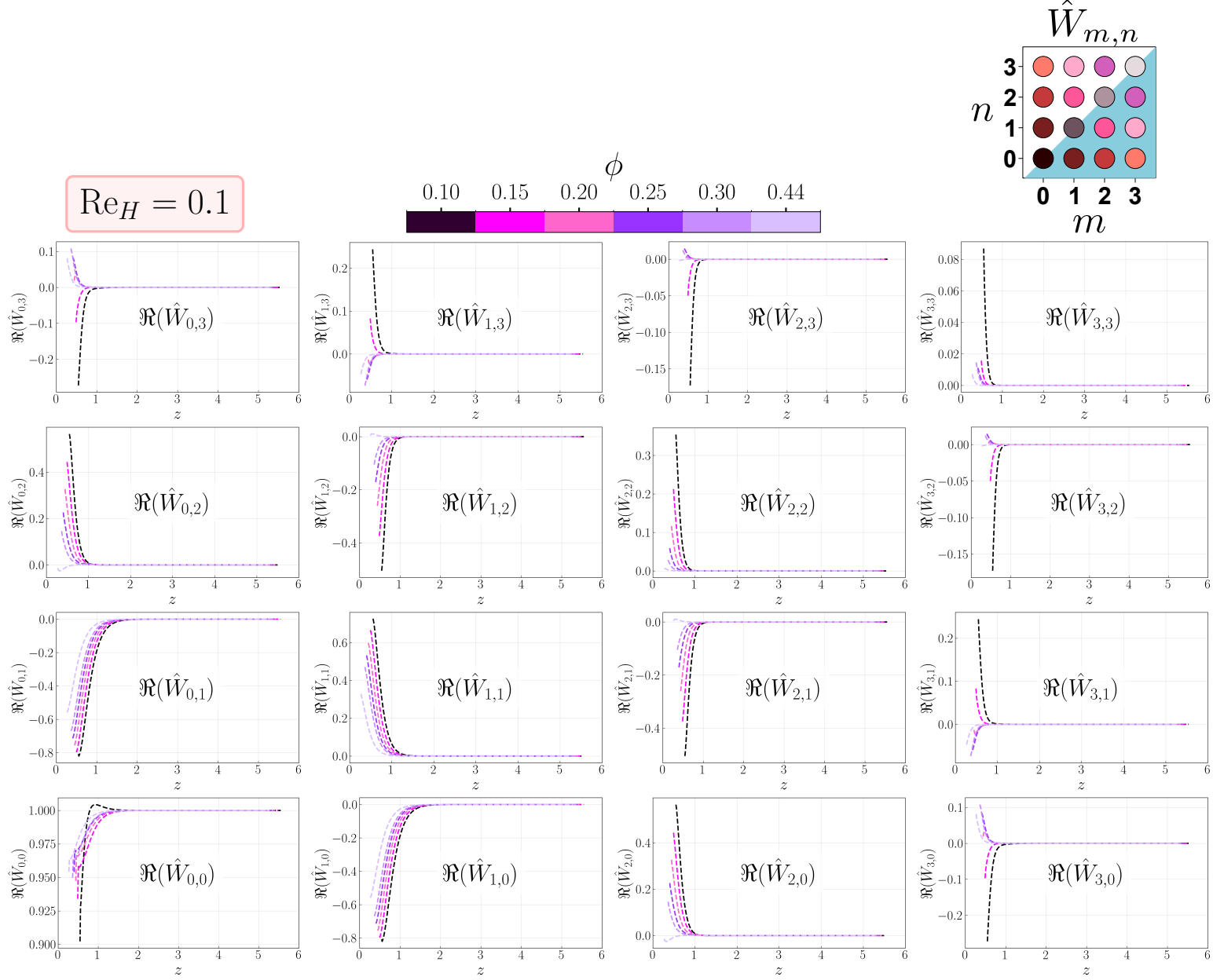


Figure B.3: Streamwise evolution of the real part $\Re(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.1$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

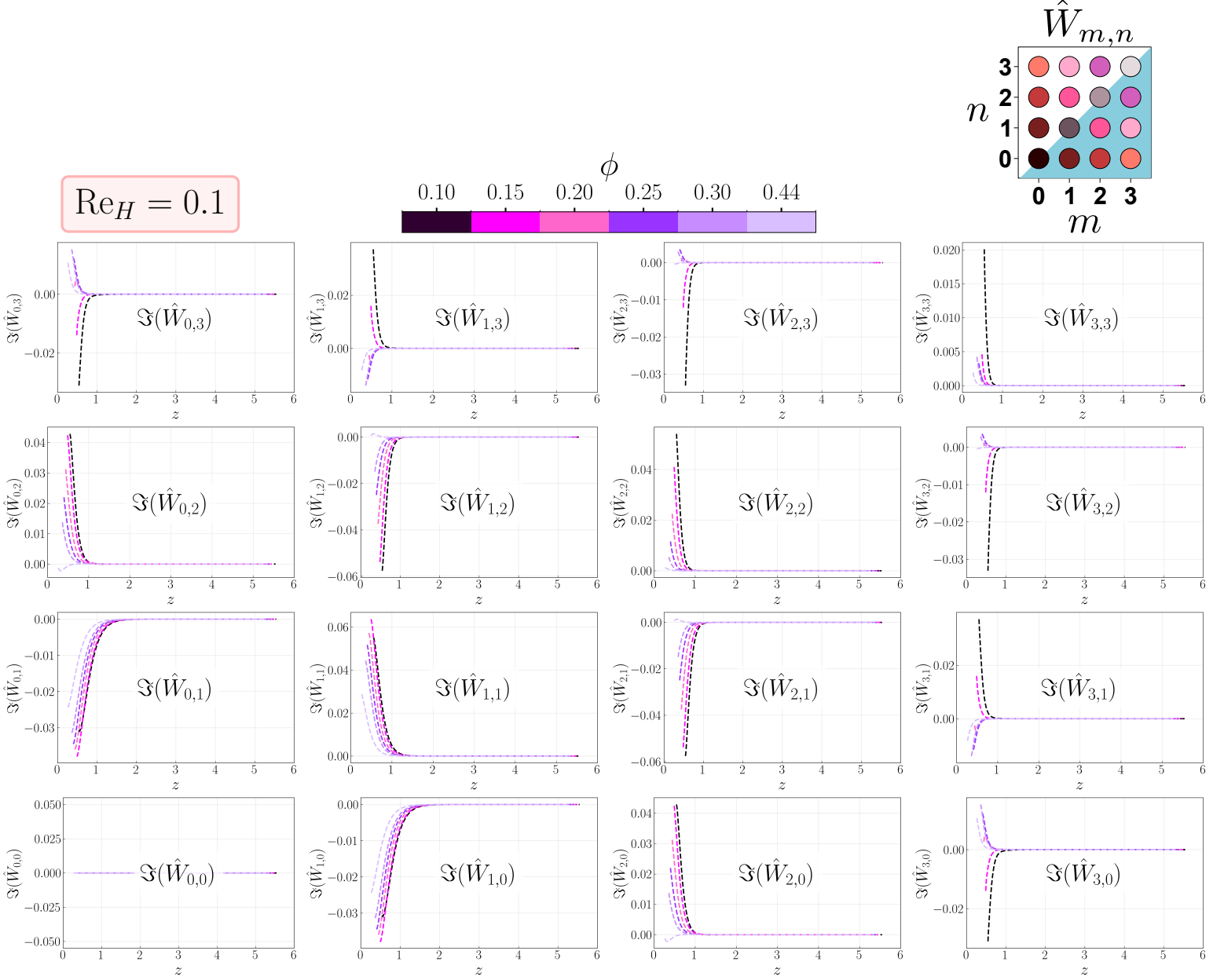


Figure B.4: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.1$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

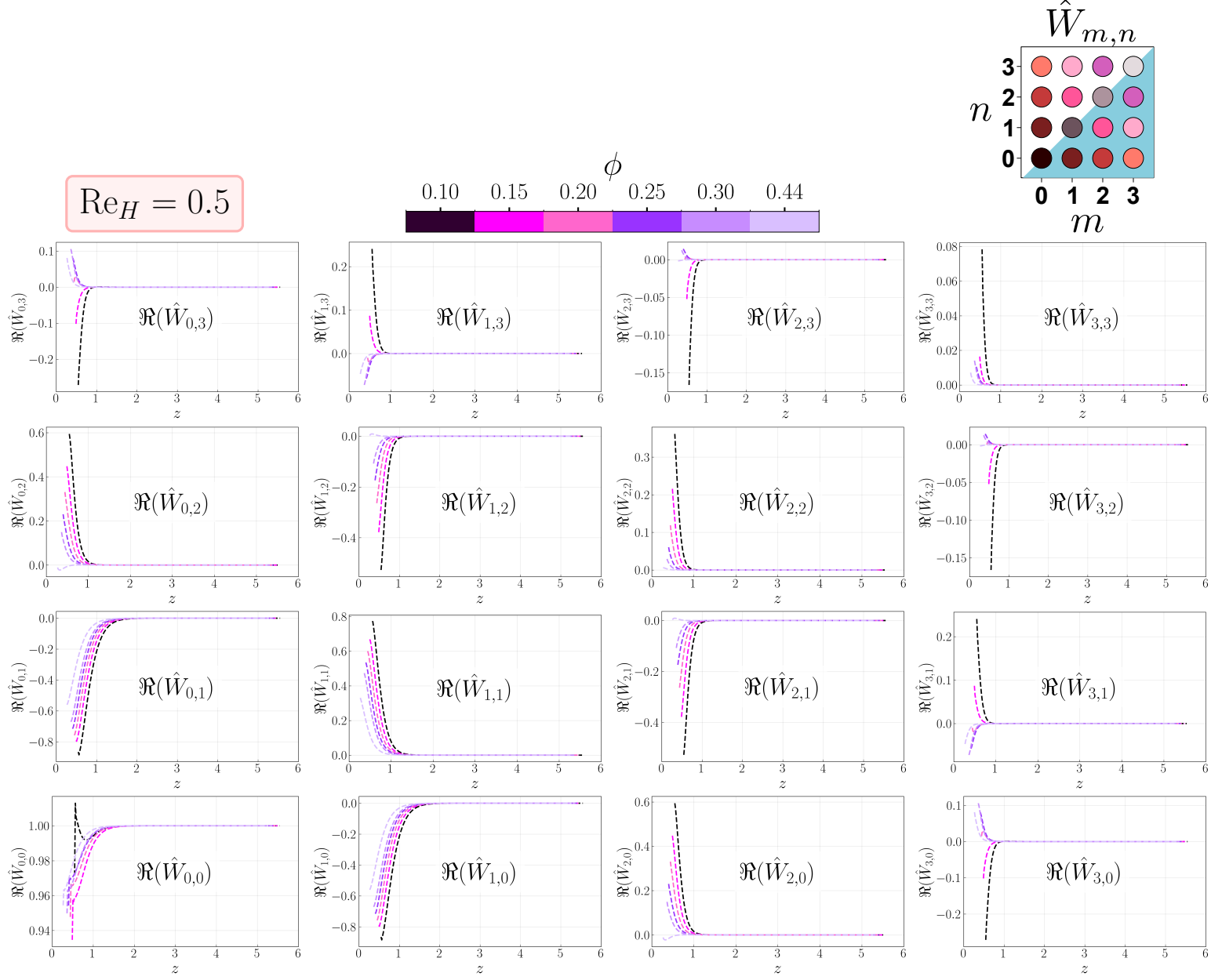


Figure B.5: Streamwise evolution of the real part $\mathcal{R}(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.5$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

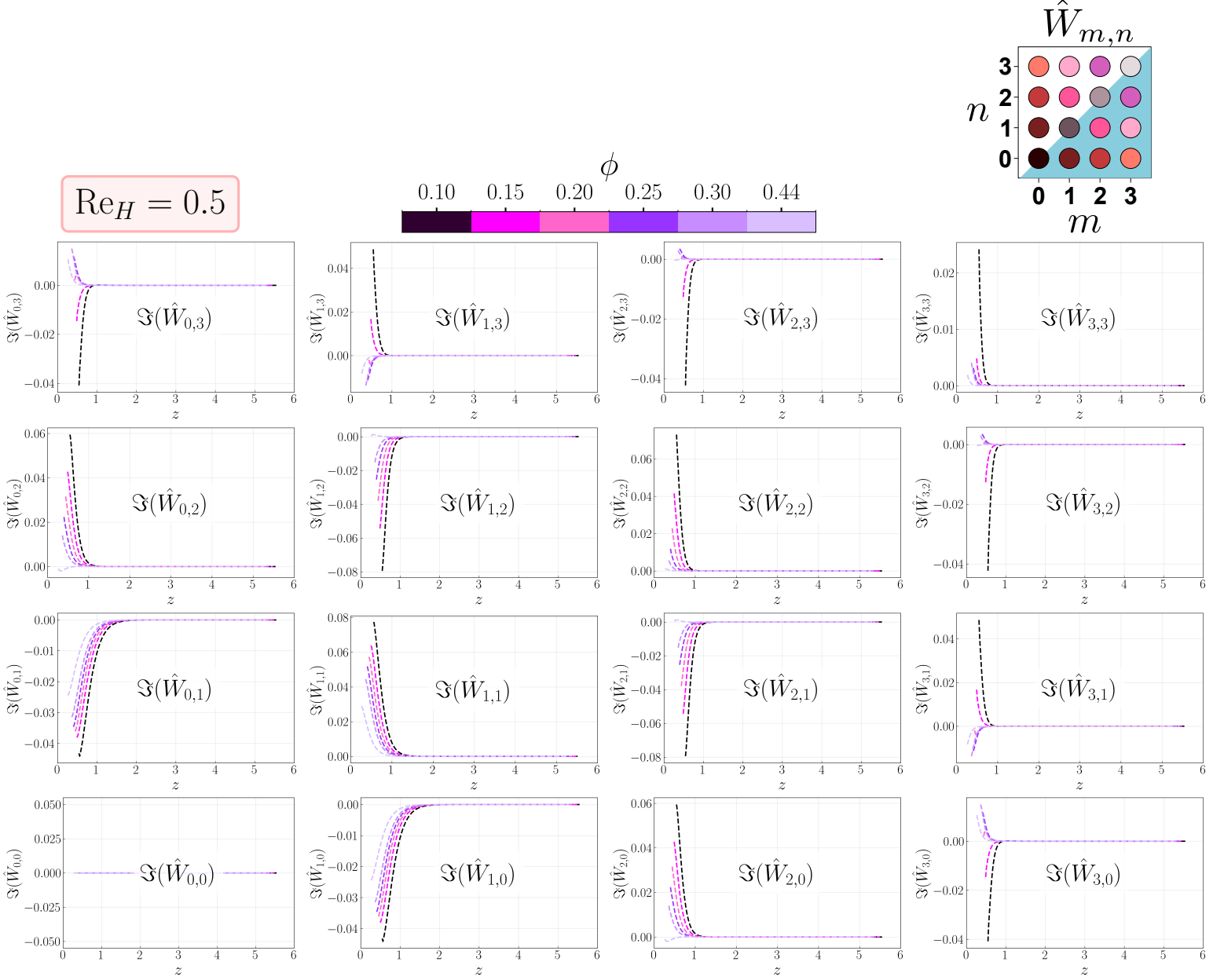


Figure B.6: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 0.5$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

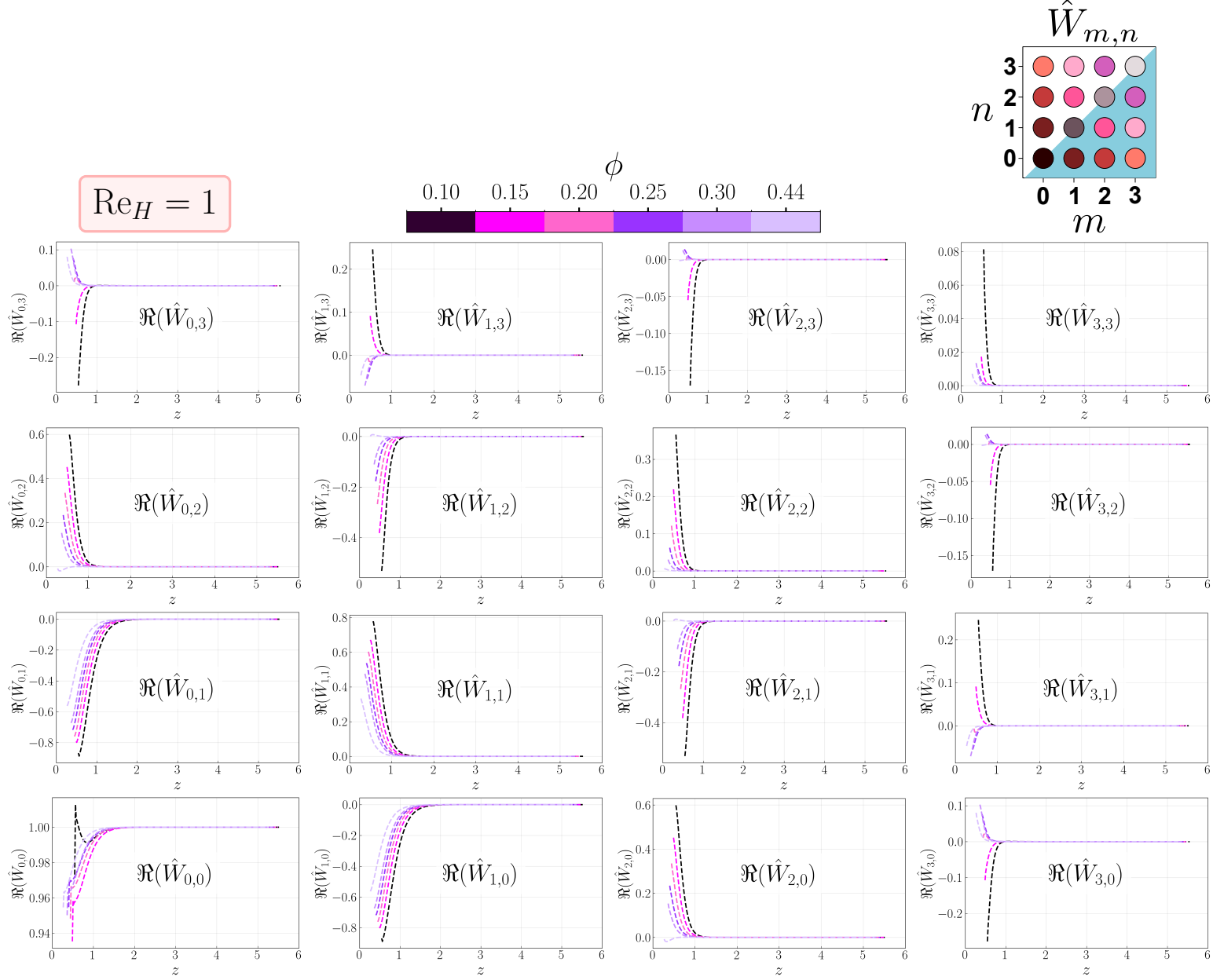


Figure B.7: Streamwise evolution of the real part $\Re(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 1$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

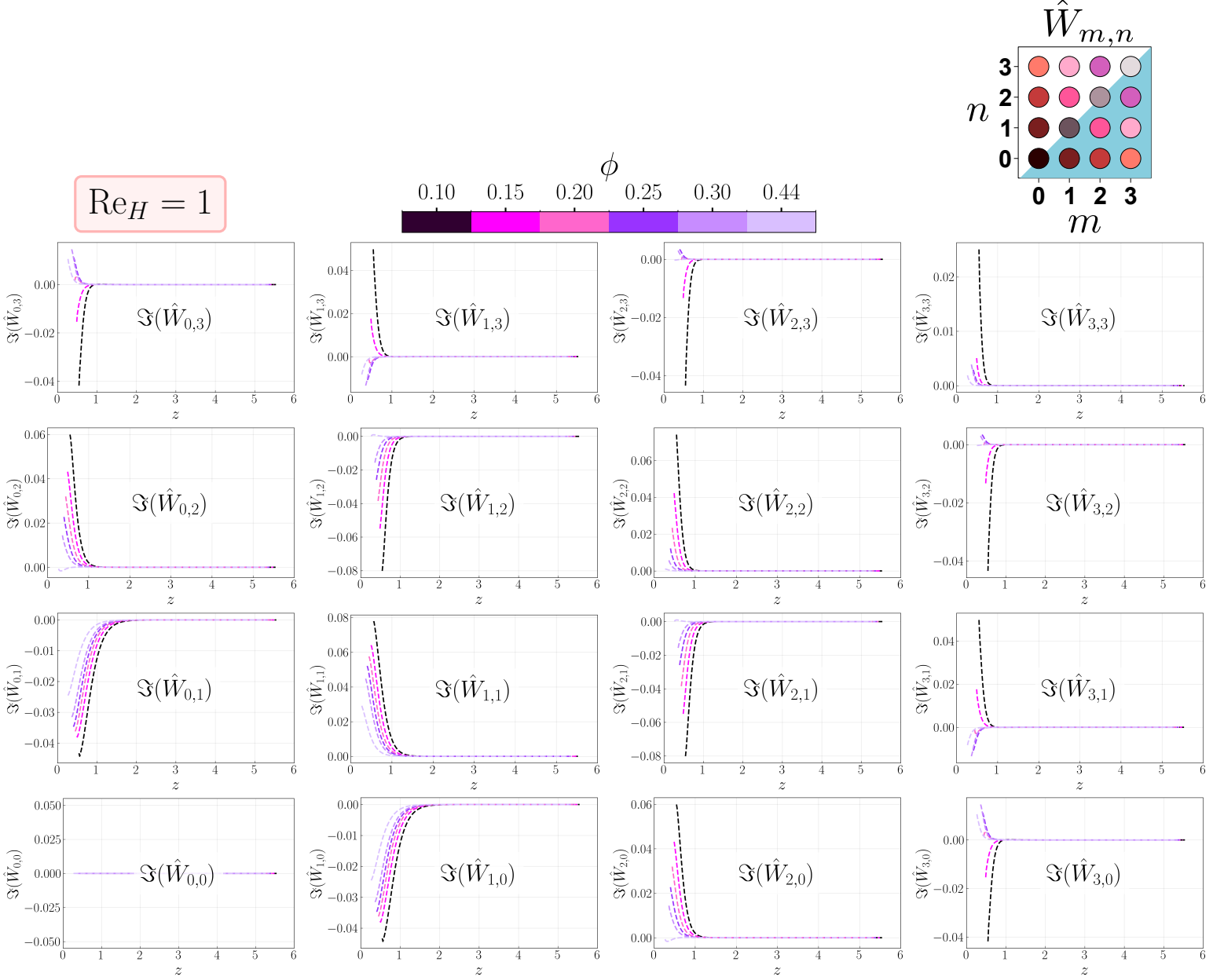


Figure B.8: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 1$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

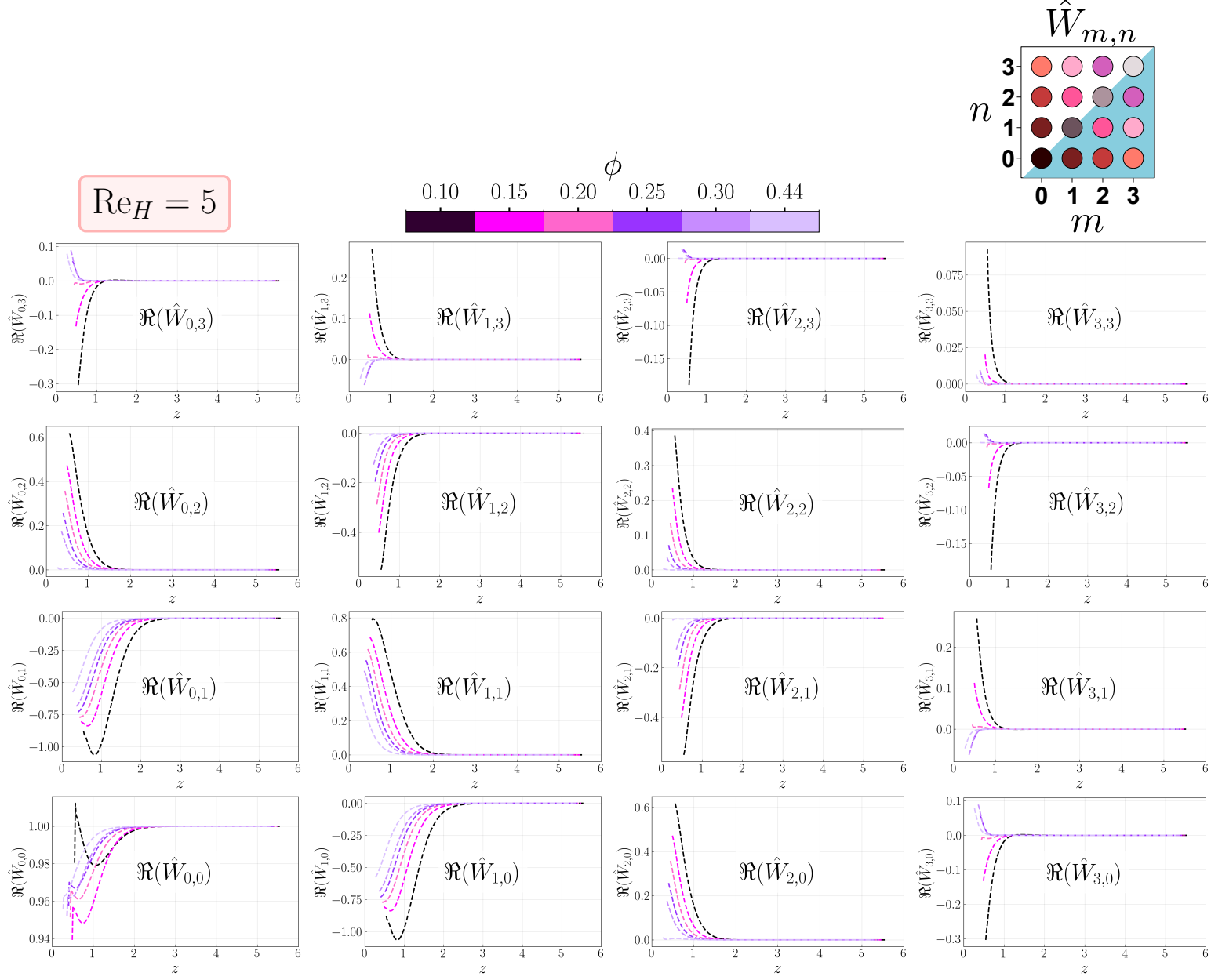


Figure B.9: Streamwise evolution of the real part $\Re(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 5$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

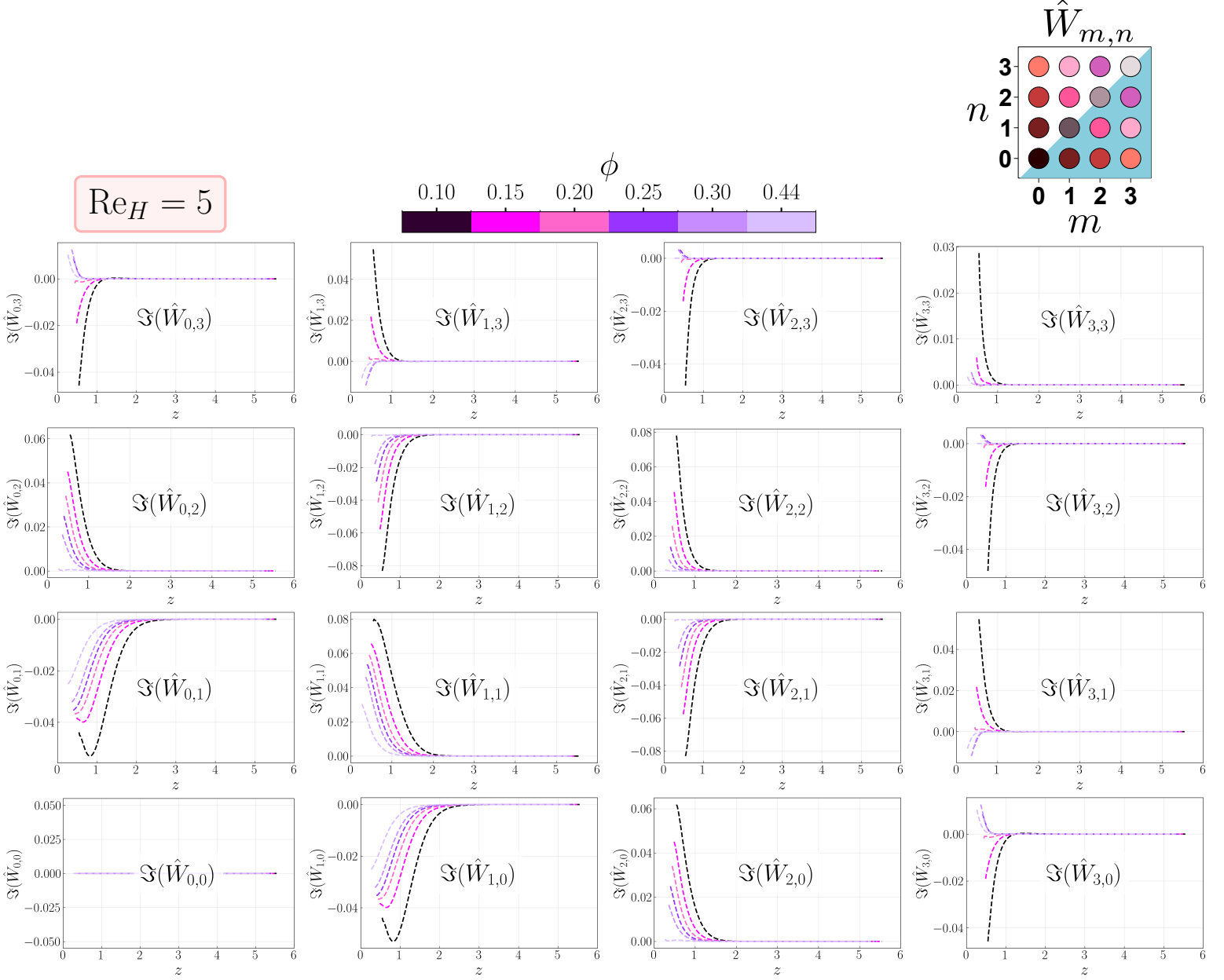


Figure B.10: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 5$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

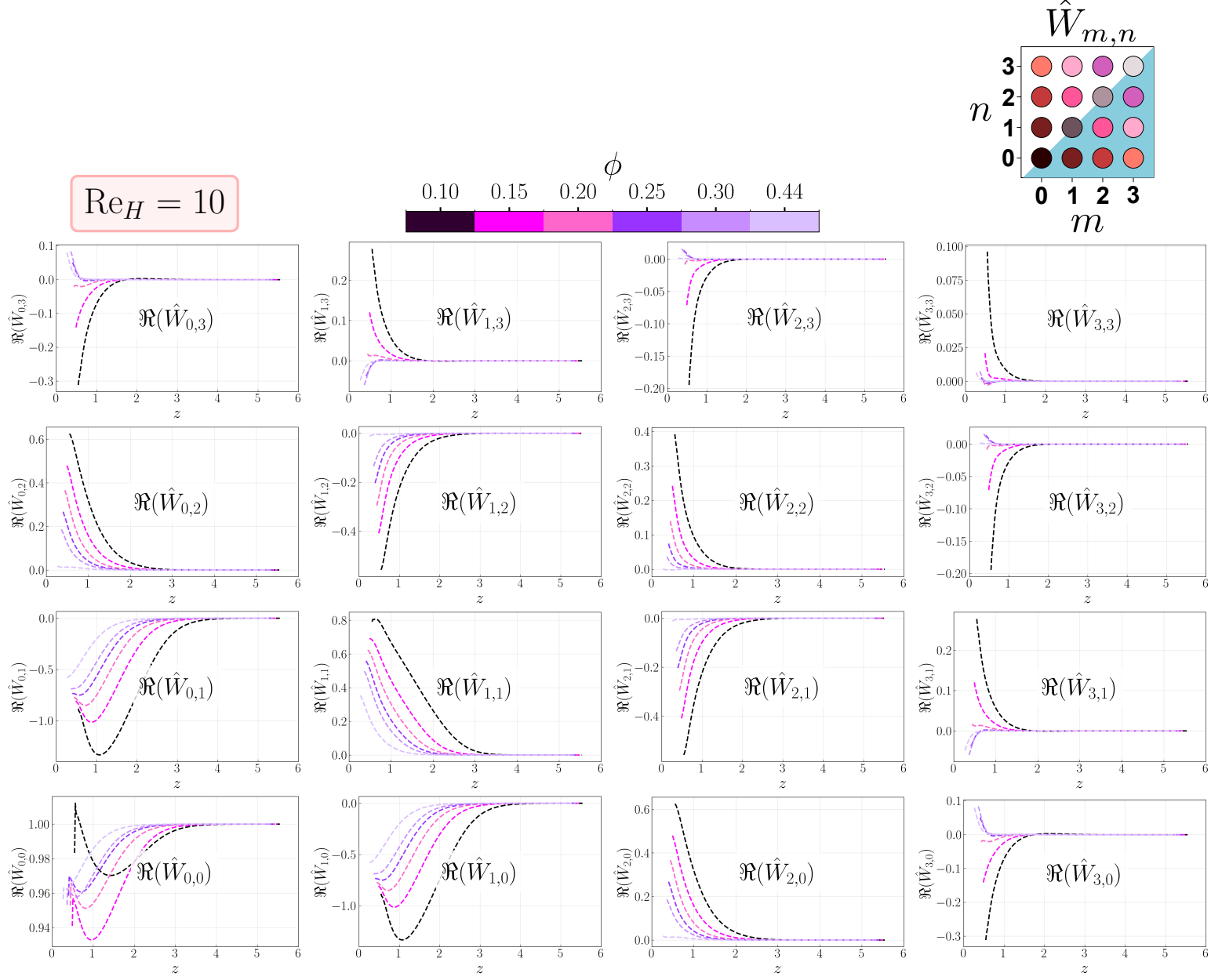


Figure B.11: Streamwise evolution of the real part $\mathcal{R}(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 10$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

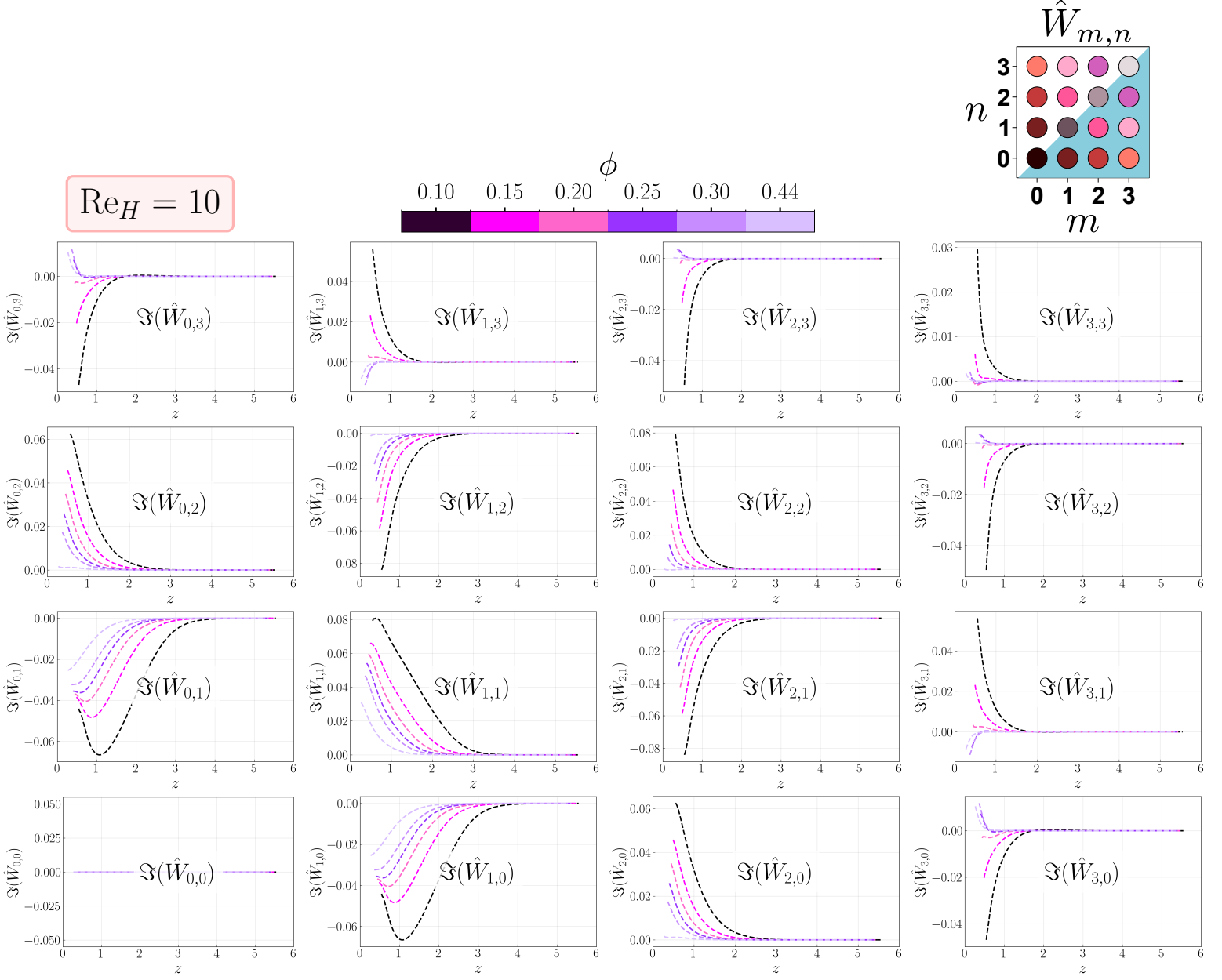


Figure B.12: Streamwise evolution of the imaginary part $\Im(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 10$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.

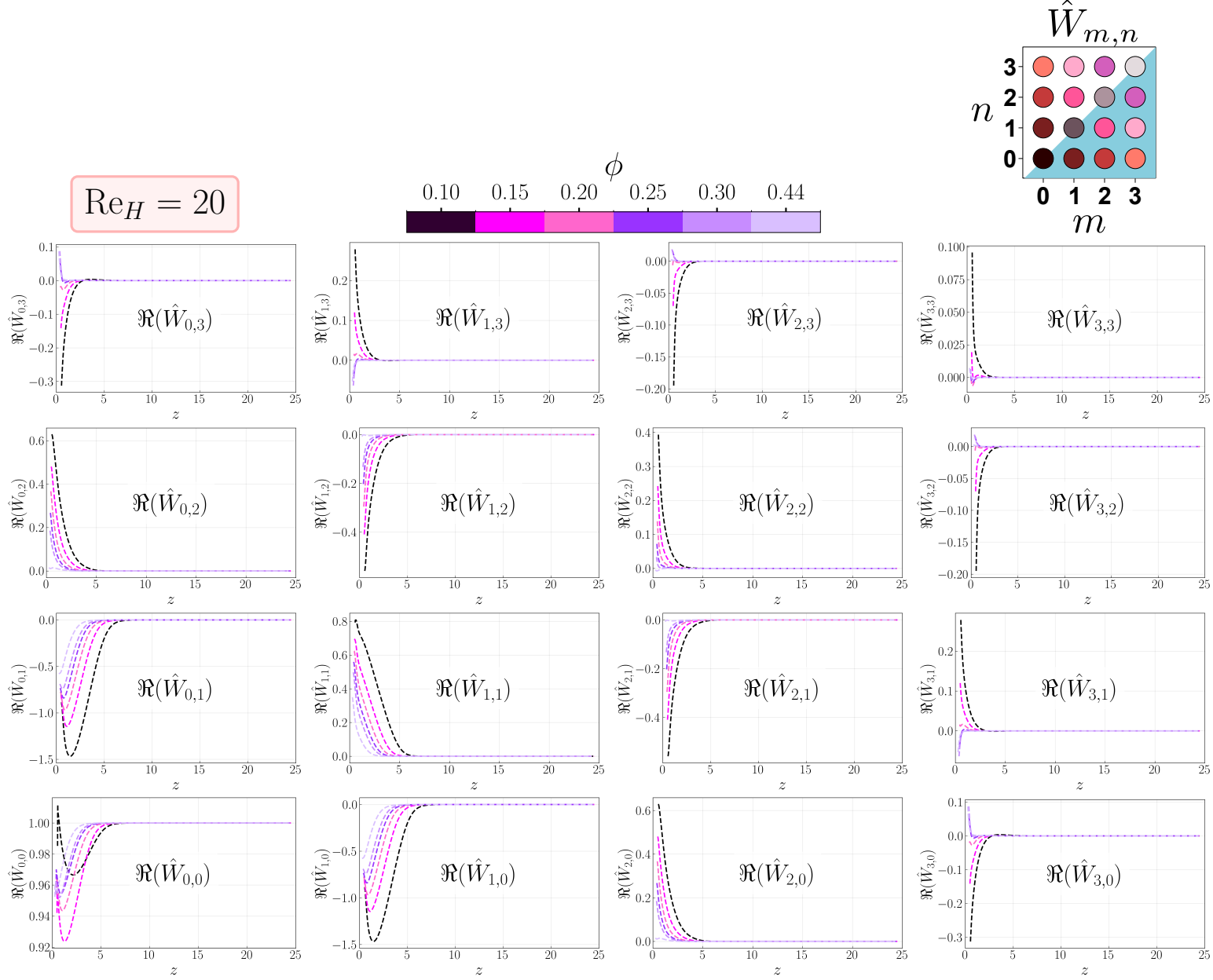


Figure B.13: Streamwise evolution of the real part $\mathcal{R}(\hat{W}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 20$. Each panel shows one mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{W}_{m,n} = \hat{W}_{n,m}$.